

# SETH: Subcode Equivalence in the Head

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## Abstract

The Multi-Party Computation in-the-Head (MPCitH) paradigm has proven to be a versatile tool to design post-quantum digital signature schemes. Recently, MPCitH was improved with Threshold Computation in-the-Head (TCitH) and Vector Oblivious Linear Evaluation in-the-Head (VOLEitH). In this paper we propose polynomial modeling for the Subcode Equivalence Problem (SEP) and derive a TCitH and a VOLEitH digital signature scheme from it. Our signatures have way better performance when compared to other SEP-based signatures and achieve similar performance when compared to other modern signatures from similar assumptions, like those based on the Permuted Kernel Problem or the Linear Code Equivalence Problem.

**Key Words:** Code-based cryptography, Subcode Equivalence, Digital Signature, VOLEitH, TCitH.

## 1 Introduction

Following the standardization of the post-quantum digital signature schemes CRYSTALS-Dilithium, FALCON, and SPHINCS<sup>+</sup>, NIST issued a call for additional post-quantum digital signature schemes in September 2022. As

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of now, 14 candidates reached the second round of this second call. Six of these leverage “In-the-Head” paradigms: *MPCitH* converts MPC protocols into ZK proofs via view commitments, *TCitH* optimizes this using threshold secret sharing to reduce signature size, and *VOLEitH* achieves public verifiability by combining VOLE correlations with all-but-one vector commitments.

**Subcode Equivalence Problem.** Given two linear codes, the Subcode Equivalence Problem (SEP) asks to find a permutation sending one of the two codes into a subcode of the other. Despite being recently shown to be NP-Complete [BGK17], SEP has seen limited application in cryptography: as of today, the only protocol based on it is the identification scheme presented in [BGK17], which is a variant of the Girault one [Gir90] and suffers from the same efficiency problems, since it requires the prover to send multiple matrices.

**Our Contribution.** Following a similar approach to [BBGK24, BMM26], we propose a polynomial modeling for SEP. Then, using the Polynomial Interactive Oracle Proof formalism, we are able to derive a TCitH and a VOLEitH signature from it. Our signatures have better performance when compared to other SEP-based signatures like the ones presented in [BGK17, BBG<sup>+</sup>26] and achieve similar performance when compared to other modern signatures from similar assumptions, like Linear Code Equivalence or PKP [BBGK24, BBB<sup>+</sup>24a, ABB<sup>+</sup>24]. Compared to the recent work [BMM26], that tackles the Linear Code Equivalence Problem (LEP) with a similar approach, our signature offers the following advantages:

- while the exact security of LEP is still uncertain [PR97, BBPS23], SEP is NP-hard, offering higher security guarantees;
- shorter keys and also a smaller field size;
- shorter and faster signatures compared to [BMM26].

## 2 Preliminaries

In this paper,  $q$  denotes a prime and  $\mathbb{F}_q$  indicates the finite field with  $q$  elements. We will use boldface uppercase letters to denote matrices, and boldface lowercase letters to denote vectors.

Let  $S$  be a set, then  $s \xleftarrow{\$} S$  denotes that  $s$  is sampled uniformly at random from the set  $S$ . If an element  $s$  is randomly sampled from  $S$  using a seed  $\text{seed}$ , we will write  $s \xleftarrow{\$, \text{seed}} S$ .

Finally, let  $A$  be an algorithm. If  $A$  is deterministic, we write  $y \leftarrow A(X_1, \dots, X_n)$  to denote that  $y$  is computed as the output of algorithm  $A$  with inputs  $X_1, \dots, X_n$ . Instead, if  $A$  is a probabilistic algorithm, we write  $y \xleftarrow{\$} A(X_1, \dots, X_n)$  to highlight the probabilistic nature of  $A$ .

For basic information about interactive proofs and zero-knowledge protocols we refer the reader to [DN06], while we refer to [HP03] for basic notions about coding theory. We recall here only the subcode equivalence problem and the concept of Polynomial Interactive Oracle Proofs.

## 2.1 Subcode Equivalence Problem

A  $[n, k]_q$  linear code  $\mathcal{C}$  is a  $k$ -dimensional subspace of  $\mathbb{F}_q^n$ :  $n$  is the length while  $k$  is the dimension of the code. A code  $\mathcal{C} \subseteq \mathbb{F}_q^n$  with dimension  $k$  can be represented through a *generator matrix*, that is, a full-rank matrix  $\mathbf{G} \in \mathbb{F}_q^{k \times n}$  whose rows form a basis for the code. An alternative but equivalent representation of  $\mathcal{C}$  is the parity-check matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$ , whose rows form a basis for the null space of the code.

**Definition 2.1** (Subcode Equivalence Problem). Given two codes  $\mathcal{C}, \mathcal{C}' \subseteq \mathbb{F}_q^n$ , defined by the parity-check matrix  $\mathbf{H} \in \mathbb{F}_q^{(n-k) \times n}$  and the generator matrix  $\mathbf{G}' \in \mathbb{F}_q^{k' \times n}$ , respectively, with  $k' < k$ , find a permutation matrix  $\mathbf{P} \in S_n$  such that  $\mathbf{G}'\mathbf{P}\mathbf{H}^\top = \mathbf{0}$ .

We recall that SEP is NP-complete [BGK17]. The hardest instances for SEP are obtained when both  $\mathbf{G}'$  and  $\mathbf{H}$  do not have repeated columns and when there is, on average, a unique solution [SBC23].

## 2.2 Polynomial Interactive Oracle Proofs

Polynomial Interactive Oracle Proofs (PIOP) [BFS20] are interactive proofs where in each round of interaction the prover provides the verifier with oracle access to a multivariate polynomial function of bounded degree. Then, the verifier may ask the evaluation of the polynomial on arbitrary points of its choice.

Let  $(\mathbf{x}, \mathbf{w})$  be an instance of a hard relation  $R$  (i.e. finding  $\mathbf{w}$  knowing only  $\mathbf{x}$  is hard), with  $\ell$  being the length of  $\mathbf{w} = (w_1, \dots, w_\ell) \in \mathbb{F}_q^\ell$ . Moreover, let  $F = \{f_1, \dots, f_t\}$  be a system of polynomial equations where  $f_i \in \mathbb{F}_{q^r}[X_1, \dots, X_\ell]$  has degree at most  $d$ , and such that  $F(\bar{\mathbf{w}}) = 0$  if and only if  $(\mathbf{x}, \bar{\mathbf{w}}) \in R$ . Finally, let us assume that the prover has a way to commit polynomials and to provably open some evaluations later while keeping hidden the other evaluations. From a high-level perspective, a PIOP is structured as follows:

1. The prover generates  $\ell$  degree-1 polynomials  $p_j(X) = w_jX + v_j$  for some random  $v_1, \dots, v_\ell \in \mathbb{F}_{q^r}$ . The prover also generates  $d-1$  additional

degree-1 polynomials  $\bar{p}_i(X) = \bar{u}_i X + \bar{v}_i$ ,  $i = 0, \dots, d-2$ ,  $\bar{u}_i, \bar{v}_i \in \mathbb{F}_{q^r}$ .<sup>1</sup> Finally, the prover commits to these polynomials and sends the commitment **com** to the verifier. This is done using GGM trees, as explained in [BBDdSG<sup>+</sup>23].

2. The verifier picks a random  $t$ -tuple  $(\gamma_1, \dots, \gamma_t) \in \mathbb{F}_{q^r}^t$  and sends it to the prover.
3. The prover computes the polynomials  $\bar{f}_i(X) = X^d f_i\left(\frac{p_1(X)}{X}, \dots, \frac{p_\ell(X)}{X}\right)$  and computes  $\bar{F} = \sum \gamma_i \bar{f}_i$ . Notice that, by construction,  $\deg \bar{F} = d-1$  since, for all  $i = 1, \dots, t$ ,  $\bar{f}_i(X) = f_i(\mathbf{w}) \cdot X^d + \dots$  and  $f_i(\mathbf{w}) = 0$ . The prover finally computes the polynomial  $F_{\text{masked}} = \bar{F} + \text{mask}(X)$ , with  $\text{mask}(X) = \sum_{i=0}^{d-2} (\bar{u}_i X + \bar{v}_i) X^i$ , and sends it to the verifier.
4. The verifier picks a random  $\Delta \in \mathbb{F}_{q^r}$  and sends it to the prover.
5. The prover sends  $s_j = p_j(\Delta)$  for all  $j = 1, \dots, \ell$  and  $\bar{s}_i = \bar{p}_i(\Delta)$  for all  $i = 0, \dots, d-2$ , together with a proof  $\pi$  proving that the evaluations are consistent with the initial commitment **com**.
6. The verifier checks  $\pi$  and that  $\deg(F_{\text{masked}}) = d-1$ . Then computes  $\bar{F}(\Delta) = \sum_{i=0}^{d-2} \bar{s}_i \Delta^i$ , and checks that  $\bar{F}(\Delta) + \text{mask}(\Delta) = F_{\text{masked}}(\Delta)$ .

It is possible to prove that the above protocol has soundness error of

$$\frac{1}{q^r} + \left(1 - \frac{1}{q^r}\right) \cdot \frac{d}{q^r},$$

When  $\mathbb{F}_{q^r}$  is a small field, this leads to a high soundness error and therefore results in a low level of security when converted into a digital signature scheme. To reduce the soundness error we have two choices:

- TCitH: The verifier samples a random matrix  $\Gamma \in \mathbb{F}_{q^r}^{\rho \times t}$  and the prover computes  $\rho$  polynomials  $F_{\text{masked},j} = \sum_{i=0}^{t-1} \Gamma_{j,i} \bar{f}_i + \text{mask}_j$ : in the verification phase, the verifier performs the same checks for each polynomial using the same evaluation point  $\Delta \in \mathbb{F}_{q^r}$ . Hence, in this case the soundness error is  $\frac{1}{q^{\rho r}} + \left(1 - \frac{1}{q^{\rho r}}\right) \cdot \frac{d}{q^r}$ , which is approximately  $d/q^r$  if  $\rho r \log_2(q) \geq \lambda$ . Then, the error is further reduced by performing  $\tau$  parallel repetitions of the whole protocol to achieve  $(d/q^r)^\tau < 2^{-\lambda}$ .
- Sample the challenge  $\Delta$  from a set having an exponential number of elements. This requires to lift the polynomials  $p_j$  and  $\bar{p}_i$  to an extension  $\mathbb{F}_{q^{r\tau}}$  of  $\mathbb{F}_q$  so that  $q^{r\tau}$  is exponentially large. This is the approach

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<sup>1</sup>In the VOLE formalism, these polynomials are called VOLE correlations.

of VOLEitH protocols and results in signatures having 7 rounds, since this lifting requires an additional round. However, VOLEitH signatures are in general shorter, since they do not incur in the security loss due to parallel repetitions [KZ20].

**Remark 2.1.** In both cases, we can consider  $f_1, \dots, f_t \in \mathbb{F}_{q^m}[X_1, \dots, X_\ell]$  with  $m \leq r$  and then lift them to an extension  $\mathbb{F}_{q^r}$ , in the case of TCitH, or  $\mathbb{F}_{q^{r\tau}}$ , in the case of VOLEitH: to achieve this, it must hold  $m \mid r$  or  $m \mid r\tau$ , respectively. Then, the protocol proceeds as described above, by defining the polynomials  $p_j(X)$  and  $\bar{p}_i(X)$  over the extension field.

In both cases, the digital signature scheme is then obtained by applying the Fiat-Shamir transform to the interactive protocol.

**Lifting to a larger fields** Directly selecting  $q^r$  to be exponentially large to achieve negligible soundness error is impractical, as it requires a GGM tree with an exponential number of leaves (see [BBDdSG<sup>+</sup>23]). To address this, we generate  $\tau$  distinct VOLE correlations over a smaller field  $\mathbb{F}_{q^r}$  and aggregate them into a single correlation over the extension field  $\mathbb{F}_{q^{r\tau}}$ . First, the prover and verifier generate  $\tau$  (small) VOLE correlations, denoted as  $(\mathbf{u}_i, \mathbf{v}_i)$ , with  $\mathbf{u} \in \mathbb{F}_q^\ell$ ,  $\mathbf{v}_i \in \mathbb{F}_{q^r}^\ell$ , and  $(\Delta_i, \mathbf{s}_i)$ , with  $\Delta_i \in \mathbb{F}_{q^r}$ ,  $\mathbf{s}_i \in \mathbb{F}_{q^r}^\ell$ . These instances are concatenated to form matrices  $\mathbf{U} \in \mathbb{F}_q^{\ell \times \tau}$  and  $\mathbf{V}, \mathbf{S}' \in \mathbb{F}_{q^r}^{\ell \times \tau}$  satisfying  $\mathbf{S}' = \mathbf{U} \cdot \text{diag}(\mathbf{\Delta}) + \mathbf{V}$ , where  $\mathbf{\Delta} = (\Delta_1, \dots, \Delta_\tau)^\top$ . To obtain a single valid correlation for a fixed witness  $\mathbf{u}$ , the prover computes and sends correction vectors  $\mathbf{e}_i = \mathbf{u}_i - \mathbf{u}$  for each instance. The verifier uses these vectors to correct their share  $\mathbf{S}'$ , deriving a final matrix  $\mathbf{S}$  such that  $\mathbf{S} = \mathbf{u} \cdot \mathbf{\Delta}^\top + \mathbf{V}$ . By interpreting  $\mathbf{\Delta}^\top$  and each row of  $\mathbf{v}$  and  $\mathbf{S}$  as elements of the extension field  $\mathbb{F}_{q^{r\tau}}$ , this relation yields a single VOLE correlation of length  $\ell$  satisfying  $\mathbf{s}^* = \mathbf{u} \cdot \mathbf{\Delta}^* + \mathbf{v}^*$  that achieves a soundness error of approximately  $2^{-\lambda}$ . Since the prover could cheat by sending incorrect values for  $\mathbf{e}_i$ , the parties must perform a VOLE consistency check [Roy22] by applying a random linear universal hash function to compress  $\mathbf{u}$  and  $\mathbf{V}$ , ensuring that the verifier's reconstructed share matches the prover's committed values.

### 3 Modeling SEP

To build a PIOP we need to find a polynomial system of equations  $F_{\text{SEP}}$  that models a given SEP instance  $(\mathcal{C}, \mathcal{C}')$  such that  $\pi$  is a solution of it if and only if  $F_{\text{SEP}}(\pi) = 0$ .

Throughout this section, we will write  $d = \lceil \log_2(n) \rceil$ . We will denote by  $\mathcal{B}^j \in \mathbb{F}_2^d$  the binary representation of an integer  $j \in [0, n-1]$ , and by  $\bar{\mathcal{B}}^j$  its bitwise complement. We recall that  $k$  and  $k'$  are the dimension of  $\mathcal{C}$  and  $\mathcal{C}'$ , while  $n$  is their length.

Observe that, given  $\mathbf{b}^1, \mathbf{b}^2 \in \mathbb{F}_2^d$ ,  $\mathbf{b}^1 \oplus \mathbf{b}^2 = (1, \dots, 1)^\top$  if and only if  $\mathbf{b}^2 = \overline{\mathbf{b}^1}$ . As a consequence, the product of the entries of  $(\mathbf{b}^1 \oplus \mathbf{b}^2)$  is equal to 1 if and only if  $\mathbf{b}^2 = \overline{\mathbf{b}^1}$ . Given a integer  $\text{pos} \in [0, n-1]$ , let  $\mathcal{B}^{\text{pos}} = (p_0, \dots, p_{d-1}) \in \mathbb{F}_2^d$ . Then, the vector  $\mathbf{z} = (z_1, \dots, z_n) \in \mathbb{F}_2^n$  defined as

$$\forall j \in [0, n-1], z_j := \prod_{k=0}^{d-1} (p_k \oplus \overline{\mathcal{B}^j_k}) \quad (1)$$

contains exactly one entry equal to 1 at position  $j = \text{pos}$ , while all its other coordinates are equal to 0. In other words, this procedure reconstructs an elementary vector  $\mathbf{z}$  given the position of the 1. With this idea, we can represent a permutation matrix  $\mathbf{P}$  by  $n$  integers  $\text{pos}_0, \dots, \text{pos}_{n-1} \in [0, n-1]$ , which correspond to the positions of the non-zero entry in each row of  $\mathbf{P}$ : then, we can reconstruct each row using the procedure shown above. We thus employ  $n \cdot d$  variables  $\{p_{i,j}\}_{i \in [0, n-1], j \in [0, d-1]}$  to represent the permutation matrix  $\mathbf{P}$  by means of  $\text{pos}_0, \dots, \text{pos}_{n-1}$  by setting  $\mathcal{B}^{\text{pos}_i} = (p_{i,0}, \dots, p_{i,d-1}) \forall i \in [0, n-1]$ . Then,  $F_{\text{SEP}}$  is realized by the following sets of polynomials:

- $n$  equations of degree  $d$  imposing that each column of  $\mathbf{P}$  has exactly one non-zero entry

$$\mathcal{S}_{\text{col}} = \left\{ \left( \sum_{i=0}^{n-1} Z(i, j) \right) - 1 = 0 \mid j \in [0, n-1] \right\} \quad (2)$$

where  $\forall i \in [0, n-1], j \in [0, n-1]$  the parameter  $Z(i, j)$  is defined as

$$Z(i, j) := \prod_{l=0}^{d-1} (p_{i,l} \oplus \overline{\mathcal{B}^j_l}). \quad (3)$$

From Equation (1) it follows that for all  $i \in [0, n-1]$  the variables  $(Z(i, 0), \dots, Z(i, n-1))$  encode the  $i$ -th row of  $\mathbf{P}$ . Indeed, for all  $i \in [0, n-1]$ ,  $Z(i, j) = 1$  if  $j = \text{pos}_i$ , whereas  $Z(i, j) = 0$  if  $j \neq \text{pos}_i$ .

- $k' \cdot (n - k)$  equations of degree  $d$  imposing that the secret  $\mathbf{P}$  satisfies the equation  $\mathbf{G}'\mathbf{P}\mathbf{H}^\top = \mathbf{0}$

$$\mathcal{S}_{\text{eq}} = \left\{ \sum_{m=0}^{n-1} \left( \mathbf{G}'_{i,m} \cdot \sum_{l=0}^{n-1} Z(m, l) \cdot \mathbf{H}_{j,l} \right) = 0 \mid i \in [0, k'-1], \right. \\ \left. j \in [0, n-k-1] \right\} \quad (4)$$

**Remark 3.1.** Overall, the sets  $\mathcal{S}_{\text{col}}, \mathcal{S}_{\text{eq}}$  form a system of  $n + k'(n - k)$  polynomial constraints of degree maximum  $d$  in  $n \cdot d$  variables.

**Remark 3.2.** The above modelling is correct only for instances defined over  $\mathbb{F}_{2^m}$ . Indeed, when working in odd characteristic there is no guarantee that the variables  $\{p_{i,j}\}$  are binary. Also, in this case, a higher degree relation is needed to implement the XOR (see [BMM26]), resulting in a less efficient signature scheme. For this reason all our proposed instances are in characteristic 2.

It is easy to see that the following holds:

**Proposition 3.1.** *Let  $(\mathcal{C}, \mathcal{C}')$  be an instance of SEP. A permutation  $\pi$  is a solution for it if and only if it satisfies the set of equations  $\mathcal{S}_{\text{col}}, \mathcal{S}_{\text{eq}}$ .*

## 4 Digital Signatures from SEP

In this section we present our digital signature scheme based on SEP. We only focus on the VOLEitH version, since it achieves the best performance in terms of signature size, while we only provide a hint about the TCitH version.

Briefly, the signature scheme is defined by the following algorithms:

- **Key Generation:** The signer picks a random code  $\mathcal{C}$  with parity-check matrix  $\mathbf{H}$  and a random permutation matrix  $\mathbf{P}$ . Then, the signer selects a subcode  $\bar{\mathcal{C}}$  of  $\mathcal{C}$ . Let  $\bar{\mathbf{G}}$  the generator matrix of  $\bar{\mathcal{C}}$ . The private key is  $\mathbf{P}$ , while the public key is  $(\mathbf{G}', \mathbf{H})$ , where  $\mathbf{G}' = \bar{\mathbf{G}}\mathbf{P}^{-1}$  as generator matrix.
- **Signature Algorithm:** The signer transforms the public key in a system of equations as per Section 3, and exploits the knowledge of  $\mathbf{P}$  to obtain a solution  $\mathbf{w}$  for it. The signature algorithm is obtained by applying the Fiat-Shamir transform to the PIOP scheme explained in Section 2.2, equipped with the consistency check of VOLE to ensure soundness.
- **Verification Algorithm:** The verification is the same of the interactive version of PIOP, with the additional check of consistency of the correction vectors.

The complete algorithm is shown in Algorithms 1 and 2, where we omit the Key Generation for space reasons and where BAVC is a batch all-but-one vector commitment scheme [BBM<sup>+</sup>25]. A description of the procedures VOLE.Commit and VOLE.Reconstruct is found in [BMM26, Algorithm 9].

The security of the signature is a straightforward consequence of the security of the PIOP model and the correctness of the polynomial modeling of Section 3.

**TCitH Version** In the TCitH version of the protocol, the signer skips the check of lines 2-4 of the signature algorithm Algorithm 1, at the price of doing parallel repetitions. Indeed, the signer produces multiple pairs  $(\mathbf{u}_i, \mathbf{v}_i)$  (line 2) and then performs lines 5-9 on each couple  $(\mathbf{u}_i, \mathbf{v}_i)$  in parallel. The verifier does the same in Algorithm 2. The size of the signature obtained using the TCitH version of the protocol is computed according to the following formula:

$$\text{SIZE} = \underbrace{\lambda}_{\text{iv}} + \underbrace{32}_{\text{ctr}} + \underbrace{\lambda T_{\text{open}} + 2\lambda\tau}_{\text{decom}_I} + \underbrace{\tau \cdot r \cdot \log_2(q)}_{\Delta} + \underbrace{\tau(n \cdot d + d \cdot \rho \cdot r \cdot \log_2(q))}_{\text{t and ZK output}}$$

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**Algorithm 1** Protocol  $\text{SETH}_{\text{Sign}}(\mathbf{m}, \mathbf{sk}, \mathbf{pk})$

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Input:

- the message  $\mathbf{m}$  to be signed;
- the public key  $\mathbf{pk} = (\mathbf{G}', \mathbf{H})$  with  $\mathbf{G}' \in \mathbb{F}_{q^m}^{k' \times n}$  and  $\mathbf{H} \in \mathbb{F}_{q^m}^{k \times n}$ ;
- the private key  $\mathbf{sk} = \mathbf{P} \in \mathcal{S}_n$  such that  $\mathbf{G}'\mathbf{P}\mathbf{H}^\top = \mathbf{0}$ .

Sign:

- 1: Compute the system of degree- $d$  equations  $\{f_1, \dots, f_t\} \in \mathbb{F}_{q^m}[X_1, \dots, X_\ell]$  from  $\mathbf{pk}$  and a solution  $\mathbf{w} \in \mathbb{F}_q^\ell$  for it from  $\mathbf{sk}$  as per Section 3. As per Remark 3.1  $\ell = nd$ .
  - 2: Sample  $\mathbf{r} \xleftarrow{\$} \{0, 1\}^\lambda$ , compute  $\text{iv} \leftarrow \text{H}(\mathbf{sk} \parallel \mathbf{pk} \parallel \mathbf{m} \parallel \mathbf{r})$  and generate VOLE correlations  $(\text{com}, \text{decom}, \mathbf{e}_1, \dots, \mathbf{e}_{\tau-1}, \mathbf{u}, \mathbf{v}) := \text{VOLECommit}(\mathbf{r}, \text{iv}, \ell + d\tau + \lceil 16/\log_2(q) \rceil)$ .  $\mathbf{v}$  is a vector over  $\mathbb{F}_{q^{r\tau}}$ , with  $m \mid r\tau \approx \lambda$ , so that  $f_1, \dots, f_t$  can be embedded in  $\mathbb{F}_{q^{r\tau}}$ .
  - 3: Set  $\text{ch}_1 \leftarrow \text{H}(\mathbf{pk} \parallel \text{com} \parallel \mathbf{e}_1 \parallel \dots \parallel \mathbf{e}_{\tau-1})$ .
  - 4: Run the VOLE consistency check of [BBB<sup>+</sup>24b] to check that all the VOLE correlations are related to the same  $\mathbf{u}$  with challenge  $\text{ch}_1$ , obtaining  $\tilde{\mathbf{u}}$  and  $\tilde{\mathbf{V}}$ .
  - 5: Embed the witness  $\mathbf{w}$  into  $\mathbf{u}$  as  $\mathbf{t} = \mathbf{u}_{[0, \ell-1]} - \mathbf{w}$
  - 6: Run a VOLE-based Zero Knowledge Proof on input  $\mathbf{sk}, \mathbf{pk}$  and VOLE correlations  $\mathbf{u}$  and  $\mathbf{v}$  to obtain the coefficients  $\tilde{a}_0, \tilde{a}_1, \dots, \tilde{a}_{d-1}$  of  $F_{\text{masked}}$ . The challenges  $(\gamma_i)_{i \in [t]}$  are computed from  $\text{ch}_2 = \text{H}(\text{ch}_1 \parallel \tilde{\mathbf{u}} \parallel \tilde{\mathbf{V}} \parallel \mathbf{t})$ .
  - 7: Set  $\text{ctr} = 0$ ,  $\text{ch}_3 := \text{H}(\text{ch}_2 \parallel \tilde{a}_0 \parallel \tilde{a}_1 \parallel \dots \parallel \tilde{a}_{d-1} \parallel \text{ctr})$  and compute  $I = (\Delta_0, \dots, \Delta_{\tau-1}) \in \mathbb{F}_{q^r}^{\tau r}$  from  $\text{ch}_3$ .
  - 8: Compute  $\text{decom}_I := \text{BAVC.Open}(\text{decom}, I, \text{iv})$  (see [BMM26, Algorithm 7]). If  $\text{decom}_I = \perp$ , increment  $\text{ctr}$  and repeat steps 5-6.
  - 9: The signature is  $\sigma := ((\mathbf{e})_{i \in [\tau-1]}, \tilde{\mathbf{u}}, \mathbf{t}, (\tilde{a}_j)_{j \in [d-1]}, \text{decom}_I, \text{ch}_3, \text{iv}, \text{ctr})$ .
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## 4.1 Performance Analysis and Parameter Choice

Table 1 reports some possible parameter choices for instantiating our signature scheme. As mentioned in Remark 3.2, we only consider instances of SEP defined over  $\mathbb{F}_{2^m}$ . To choose parameters, we studied the security of subcode equivalence instances using KMP [KMRP19], SBC [SBC23] and



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**Algorithm 2** Protocol  $\text{SETH}_{\text{Verify}}(\sigma, \mathbf{m}, \mathbf{pk})$ 


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Input:

- the signature  $\sigma := ((\mathbf{e})_{i \in [\tau-1]}, \tilde{\mathbf{u}}, \mathbf{t}, (\tilde{a}_j)_{j \in [m-1]}, \text{decom}_I, \mathbf{ch}_3, \text{iv}, \text{ctr})$ ;
- the message  $\mathbf{m}$  to be signed
- the public key  $\mathbf{pk} = (\mathbf{G}', \mathbf{H})$ .

Verify:

- 1: Compute the system of degree- $d$  equations  $\{f_1, \dots, f_t\} \in \mathbb{F}_{q^m}[X_1, \dots, X_\ell]$  from  $\mathbf{pk}$ .
  - 2: Compute  $I = (\Delta_0, \dots, \Delta_{\tau-1}) \in \mathbb{F}_{q^r}^\tau$  from  $\mathbf{ch}_3$  and recover the VOLE correlation  $(\overline{\mathbf{com}}, \mathbf{S}) \leftarrow \text{VOLE.Reconstruct}(I, \text{decom}_I, \mathbf{e}_1, \dots, \mathbf{e}_{\tau-1}, \text{iv}, \ell + d \cdot r\tau + \lceil 16/\log_2(q) \rceil)$ .
  - 3: Set  $\overline{\mathbf{ch}}_1 \leftarrow \text{H}(\mathbf{pk} \parallel \overline{\mathbf{com}} \parallel \mathbf{e}_1 \parallel \dots \parallel \mathbf{e}_{\tau-1})$ .
  - 4: Run the VOLE consistency check of [BBB<sup>+</sup>24b] with challenge  $\mathbf{ch}_1$ , obtaining  $\tilde{\mathbf{S}}$ . Recompute  $\tilde{\mathbf{V}} = \tilde{\mathbf{S}} - \tilde{\mathbf{u}} \cdot [1 \mid \dots \mid 1] \cdot \text{diag}(\mathbf{\Delta})$ .
  - 5: Compute the evaluation  $\tilde{c} = F_{\text{masked}}(\mathbf{\Delta})$  from the VOLE correlations, computing the challenges  $(\gamma_i)_{i \in [t]}$  from  $\overline{\mathbf{ch}}_2 = \text{H}(\mathbf{ch}_1 \parallel \tilde{\mathbf{u}} \parallel \tilde{\mathbf{V}} \parallel \mathbf{t})$ . Recompute  $\tilde{a}_0^* = \tilde{c} - \sum_{k=1}^{d-1} \tilde{a}_k (\mathbf{\Delta}^*)^k$ , where  $\mathbf{\Delta}^*$  is the embedment of  $(\Delta_0, \dots, \Delta_{\tau-1})$  in  $\mathbb{F}_{q^{r\tau}}$ .
  - 6: Compute  $\overline{\mathbf{ch}}_3 := \text{H}(\overline{\mathbf{ch}}_2 \parallel \tilde{a}_0^* \parallel \tilde{a}_1 \parallel \dots \parallel \tilde{a}_{d-1} \parallel \text{ctr})$ .
  - 7: Accept the signature if  $\overline{\mathbf{ch}}_3 = \mathbf{ch}_3$ . Otherwise, reject.
- 

the more recent approach in [BDP26]. The expected cost of the attacks for each parameter choice is reported in the table. The signature sizes are then computed according to the following formula

$$\text{SIZE} = \underbrace{128}_{\text{iv}} + \underbrace{32}_{\text{ctr}} + \underbrace{\tau \cdot (\ell + d\lambda + 16)}_{\substack{\mathbf{t}, \text{ZK output, VOLE corrections}}} + \underbrace{\lambda T_{\text{open}} + 2\lambda\tau}_{\text{decom}_I} + \underbrace{\lambda}_{\mathbf{\Delta}} \quad (5)$$

| CODE PARAMETERS |     |      |     | ATTACKS' COSTS |        |         | SIZES (B) |         |        |
|-----------------|-----|------|-----|----------------|--------|---------|-----------|---------|--------|
| $m$             | $n$ | $k'$ | $k$ | KMP            | SBC    | [BDP26] | VOLE      | TCitH   | pk     |
| 4               | 74  | 3    | 36  | 209.62         | 204.34 | 130.31  | 3986.25   | 5287.51 | 106.5  |
| 2               | 95  | 8    | 38  | 277.51         | 225.57 | 128.56  | 4188.38   | 5658.88 | 174.0  |
| 2               | 86  | 9    | 42  | 241.35         | 203.26 | 131.65  | 4101.75   | 5500.62 | 173.25 |
| 2               | 78  | 10   | 43  | 207.95         | 179.72 | 130.77  | 4024.75   | 5358.89 | 170.0  |
| 2               | 76  | 11   | 44  | 200.74         | 172.64 | 130.63  | 4005.5    | 5323.25 | 178.75 |

Table 1: Signature sizes for  $\lambda = 128$ .

In Table 1, for each set of parameters we provide a comparison between a signature obtained using VOLEitH and another employing TCitH. In both cases, the extension degree  $r$  is set to  $r = 12$ . The PIOP parameters for the two signatures are as follows:

| SCHEME                               | SIGNATURE SIZE (B) | PUBLIC KEY (B) |
|--------------------------------------|--------------------|----------------|
| This work (VOLE)                     | 3986               | 106            |
| This work (TCitH)                    | 5287               | 106            |
| PKP [BBGK24]                         | 3600               | 100            |
| PERK (small)                         | 3480               | 100            |
| PERK (fast)                          | 4320               | 100            |
| LESS 252-192                         | 2289               | 13940          |
| LESS 252-45                          | 1313               | 97484          |
| LESS VOLE [BMM26]                    | 49877              | 13940          |
| Matrix Subcode [BBG <sup>+</sup> 26] | 4788               | 255            |

Table 2: Comparison with other signature schemes based on similar problems.

- For VOLEitH, we set  $\tau = 11$ ,  $T_{\text{open}} = 102$ . To obtain  $\lambda$ -bit challenges using  $\tau = 11$ , we use the same optimizations as in [BBB<sup>+</sup>24b].
- For TCitH, we set the repetition parameters  $\rho = 11$ ,  $\tau = 14$  and  $T_{\text{open}} = 160$ .

Table 2 shows a comparison between our protocol (with  $(\ell, m, n, k) = (3, 4, 74, 36)$ ) and other similar protocols that are participating in the NIST competition. All the considered instances fall in the NIST-I security category. As we can see, our signature sizes are competitive with other protocols and smaller when compared to the Matrix Subcode Equivalence variant [BBG<sup>+</sup>26]. Notice that, while the obtained signatures are slightly bigger than in LESS, our protocol is expected to be faster, since it achieves a lower computational complexity. Indeed, in LESS the verifier needs to perform Gaussian Elimination, which has a cost of roughly  $O(n^3)$  operations, whereas our protocol has complexity  $O(n^2 \log^2 n)$ . Moreover, our protocol features substantially smaller keys compared to LESS.

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